MODULE 08 · SECTION 1 - DESIGNING DATA PROCESSING SYSTEMS

Migrating Databases and Warehouses

Database Migration Service, Datastream, BigQuery Data Transfer Service, and moving the SQL as well as the data.

SECTION 1 of 5 EXAM WEIGHT 22% MODULE 8 of 45

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Moving a database is easy. Moving it **while the business keeps writing to it** is the job.

Larkspur's databases

SYSTEM	TODAY	WHERE IT GOES
Billing	Oracle 19c, 6 TB, 410 tables	50 tables copied into BigQuery
Tariff service	PostgreSQL on a VM	Cloud SQL for PostgreSQL
Settlement history	Hive on the Hadoop cluster	Cloud Storage, then BigQuery
Analyst SQL	Hundreds of Hive scripts	Translated to GoogleSQL

THE QUIRKS

Oracle stores dates as text, and call-centre notes hold pasted bank details.

One-time or continuous?

ONE-TIME

Dump, load, switch

- A point-in-time copy
- Writes stop during the copy
- Downtime lasts the whole copy
- Fine for small or quiet systems

CONTINUOUS

Copy, then follow changes

- A full load first
- Then every change, as it happens
- Cut over once both sides match
- Downtime of minutes, not hours

THE SIGNAL

'Minimal downtime' always means continuous replication, usually called change data capture.

Four ways to move a database

TOOL	MOVES DATA INTO	SIGNAL IN THE QUESTION
Database Migration Service	Cloud SQL or AlloyDB	Lift the database, minimal downtime
Datastream	BigQuery or Cloud Storage	Keep analytics fed from a live database
BigQuery Data Transfer Service	BigQuery	Scheduled loads, warehouses, SaaS
Batch exports	Cloud Storage, then a load job	One-off history, no change feed

THE TIEBREAKER

Ask where the data lands. That alone rules out half the options.

Database Migration Service

- **Same engine**: MySQL, PostgreSQL or SQL Server into Cloud SQL, or PostgreSQL into AlloyDB.
- **Different engine**: Oracle or SQL Server into PostgreSQL, on Cloud SQL or AlloyDB.
- A **conversion workspace** converts the schema, with Gemini helping on the hard parts.
- **Continuous** jobs load, follow changes, then **promote** the replica at cutover.

> THE TRAP
>
> It lands in Cloud SQL or AlloyDB. Not BigQuery, and not Spanner, which has its own migration tool.

Connecting to the source

METHOD	HOW TRAFFIC FLOWS	GOOGLE'S ADVICE
IP allowlist	Over the public internet	Easiest, least secure
Reverse SSH tunnel	Through a bastion VM you run	Short-lived tests, small databases
VPC peering	Private, over VPN or Interconnect	Long-running or high-volume moves
Private Service Connect	Private, through a network attachment	Easiest private option, existing instance

ON THE EXAM

A large migration over weeks, on a private link, points to VPC peering.

Datastream: change data capture as a service

- **Serverless CDC** from MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, Spanner and Salesforce.
- Lands in **BigQuery** or **Cloud Storage**. Other targets go through Dataflow templates.
- **Backfill** copies existing rows first, then changes stream on top.
- Pick **schemas, tables and even columns**. Cleared columns are never copied.

> HOW IT WORKS
>
> Datastream reads the database's change log, so the source barely notices.

Merge or append-only?

MERGE

A live mirror of the source

- Tables look like the source
- Max staleness: freshness against cost
- Needs a primary key
- Right for current-state reporting

APPEND-ONLY

Every change, kept

- One row per change event
- A change type on every row
- Works without a primary key
- Right for history and audit

THE TRADE-OFF

Lower staleness means fresher data and more merge cost. Zero minutes is freshest and dearest.

Fixing Oracle's text dates, once

```sql
-- Billing lands in raw via Datastream. Notes was never streamed.
CREATE OR REPLACE VIEW curated.bills AS
SELECT
  bill_id,
  account_id,
  PARSE_DATE('%d/%m/%Y', period_start) AS period_start,
  amount_gbp,
  TIMESTAMP_MILLIS(datastream_metadata.source_timestamp) AS changed_at
FROM raw.billing_bills;
```

6 Day first, stated explicitly. A malformed date fails loudly.

8 When the change happened in Oracle, from Datastream's metadata.

THE PATTERN

Datastream copies what is there. Cleaning belongs downstream, written once and tested.

Datastream: the fine print

- Oracle: **LogMiner** may replicate some uncommitted changes. The **binary reader** is Preview.
- No primary key? Merges use Oracle's **ROWID**, which might not be unique.
- **Private connectivity** uses VPC peering, reaching on-premises over VPN or Interconnect.
- Merges run as **BigQuery jobs**. Google recommends a slot reservation for predictable cost.

> FOR LARKSPUR
>
> Larkspur's finance director wanted predictable spend. CDC merges belong in the reservation plan.

BigQuery Data Transfer Service

- **Scheduled, managed loads** into BigQuery. No pipelines to write.
- **Warehouses**: Teradata, Amazon Redshift, Snowflake, and Hive metastore tables.
- **Databases and SaaS**: Oracle, MySQL, PostgreSQL, Salesforce, Google Ads and more.
- **Storage**: Cloud Storage, Amazon S3 and Azure Blob, on a schedule.

> THE TIEBREAKER
>
> Data Transfer Service is scheduled batch. Changes within minutes, with history, is Datastream.

The BigQuery Migration Service toolkit

TOOL	WHAT IT DOES
Migration assessment	Inventories the old warehouse, and estimates effort and cost
Batch SQL translator	Translates thousands of scripts at once into GoogleSQL
Interactive SQL translator	One query at a time, side by side
Gemini-enhanced rules	Customise translations in plain language. Preview
Data Validation Tool	Open source. Proves source and target tables match

THE POINT

Moving the data is the easy half. The SQL, and the trust in its numbers, is the hard half.

Hive to BigQuery, for Larkspur

- The files are already in **Cloud Storage**, thanks to module 07.
- **Load** them into native BigQuery tables, or **query in place** as Lakehouse tables.
- Hive scripts go through the **batch SQL translator**. Hive is a supported dialect.
- Spark jobs can stay Spark, on **Managed Service for Apache Spark**.

> THE PATTERN
>
> Migrate in waves by use case, not all at once. Each wave is validated before the next.

Validate before you cut over

- **Counts first**: rows per table, and per partition for big tables.
- **Aggregates next**: sums, minimums and maximums of key columns.
- **Row hashes** where every value matters, such as billing amounts.
- **Run in parallel** for a while, and compare the reports people actually use.

> THE TRAP
>
> Row counts alone miss a swapped day and month. Aggregates and hashes catch it.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Minimal downtime, into Cloud SQL or AlloyDB	Database Migration Service, continuous
Oracle to PostgreSQL	Database Migration Service, conversion workspace
Live database to **BigQuery**, fresh within minutes	Datastream
Keep every change for audit	Datastream, append-only
Teradata, Redshift, Snowflake, scheduled loads	BigQuery Data Transfer Service
Thousands of scripts to convert	Batch SQL translator

THE PATTERN

Destination first, then downtime, then freshness.

> Analysts need 50 of Larkspur's Oracle billing tables in BigQuery, never more than a few minutes behind the source.
>
> Oracle keeps running billing, and its database team will not accept extra query load. Nobody wants to write or run pipeline code.

What should the team use?

A. Nightly Data Pump exports to Cloud Storage, loaded by a scheduled job

B. A Database Migration Service continuous job from Oracle into BigQuery

C. Datastream from Oracle into BigQuery, with only the 50 tables selected

D. A Dataflow JDBC pipeline that queries every table every five minutes

◆ PAUSE AND DECIDE

What should the team use?

A Nightly Data Pump exports

Up to a day behind, not a few minutes.

B Database Migration Service

It lands in Cloud SQL or AlloyDB. BigQuery is not a destination.

C Datastream, 50 tables

Reads the change log, so Oracle feels little load. Minutes fresh, no code.

D A Dataflow JDBC pipeline

Code to own, and repeated full queries that Oracle's team refused.

THE SIGNAL

'A few minutes behind' plus 'no extra load' means log-based CDC. Into BigQuery, that is Datastream.

> Larkspur's tariff service runs PostgreSQL on a VM in Bristol. It must move to Cloud SQL for PostgreSQL with no more than a few minutes of downtime.
>
> The team wants weeks of testing against the copy before cutover. The Partner Interconnect is live.

How should they migrate?

A Take a full dump on a quiet Sunday night, then restore it into Cloud SQL

B A continuous Database Migration Service job over VPC peering, then promote

C Datastream from PostgreSQL straight into the new Cloud SQL instance

D A one-time Database Migration Service job, connected by IP allowlist

◆ PAUSE AND DECIDE

How should they migrate?

A. A Sunday dump and restore

Writes stop for the whole dump and restore. That is hours, not minutes.

B. Continuous, VPC peering

Follows changes for weeks of testing, then cuts over in minutes.

C. Datastream into Cloud SQL

Datastream lands in BigQuery or Cloud Storage, not Cloud SQL.

D. One-time job, IP allowlist

One-time means downtime, and the public internet ignores the Interconnect.

THE SIGNAL

'Weeks of testing' plus 'minutes of downtime' means a continuous job. Long and private means VPC peering.

> A retailer is moving a 300 TB Teradata warehouse to BigQuery. It has about 40,000 SQL scripts, and leadership wants an effort estimate before approving the project.
>
> Finance will only sign off once migrated tables are shown to match the originals.

Which set of tools fits?

A. Assessment, batch translator, Data Transfer Service, Data Validation Tool

B. Gemini in BigQuery to rewrite each script, then Storage Transfer Service

C. Datastream from Teradata into BigQuery, then Dataform assertions

D. CSV exports on a Transfer Appliance, scripts rewritten by hand, counts

◆ PAUSE AND DECIDE

Which set of tools fits?

A The Migration Service toolkit

Estimate first, translate in bulk, load on a schedule, then prove it matches.

B Gemini rewrites, Storage Transfer

Forty thousand scripts one by one, and files that never become tables.

C Datastream, then assertions

Datastream does not read Teradata, and assertions test rules, not matches.

D Appliance, hand rewrites, counts

Months of manual SQL, and row counts alone will not satisfy finance.

THE SIGNAL

Estimate, translate in bulk, move, prove. Each phrase maps to one tool in the Migration Service toolkit.

> Larkspur is setting up the Datastream stream for billing. The Oracle accounts table has a free-text notes column where call-centre agents have pasted IBANs.
>
> Analysts never use the notes. Security says bank details must not reach BigQuery in any form.

What should the team do?

A Stream every column, then hide notes from analysts with a policy tag

B Stream every column, and run a weekly scan to find and delete the IBANs

C Ask the call centre to stop pasting bank details, then stream everything

D Clear the notes column in the stream, so it is never copied to BigQuery

◆ PAUSE AND DECIDE

What should the team do?

A Stream, then a policy tag

Hidden is not absent. The IBANs still sit in BigQuery.

B Stream, then a weekly scan

Bank details live in BigQuery for up to a week at a time.

C Ask the call centre

Good practice, but years of pasted IBANs are already in the column.

D Clear the notes column

The column never leaves Oracle, and analysts lose nothing they use.

THE SIGNAL

'Must not reach BigQuery in any form' rules out every fix that happens after loading.

Module 08 in five lines

- **Minimal downtime** means continuous replication: full load, follow changes, then cut over.
- **Database Migration Service** lands in Cloud SQL or AlloyDB. Never BigQuery.
- **Datastream** feeds BigQuery from a live database. Merge for now, append-only for history.
- **Data Transfer Service** is scheduled batch. The **translators** move the SQL.
- **Prove it matches** with aggregates and hashes. Counts alone miss swapped dates.

COMING UP

Next: module 09, designing for portability and multi-cloud.